Md Shafiul Alam

Oulu, Finland mdshafiofalam.github.io 0009-0002-0344-2990 Portfolio

Education

2024-2026 **M. Sc. in Information Processing Science** Oulu, Finland

- Faculty of Information Technology and Electrical Engineering, University of Oulu
- **CGPA:** *4.88 / 5.00* (4.97, without Finnish language course)
- **Orientation:** *Information Systems*
- **Specialization:** Human-Centered Design & User Experience, Persuasive Design & Gamification
- **Thesis**: *Making Friendship with Data Centers: Understanding Community Needs for Sustainability,* **Grade:** *5 / 5*, **Supervisor:** Dr. Marianne Kinnula
  Reviewer: Dr. Dorina Rajanen, Dr. Marianne Kinnula, Dr. Netta Iivari

2017-2022 **B. Sc. in Computer Science & Engineering** Dhaka, Bangladesh

- Department of Computer Science & Engineering, Daffodil International University
- **CGPA:** *3.92 / 4.00*

Research Experience

Masters Thesis

IT Infrastructure and Community Sustainability

- Examined critical IT infrastructure (data center) operations for community sustainability through a qualitative study using the Sustainable HCI lens.
- Collected semi-structured, participatory design inputs from Oulu city residents and applied thematic analysis.
- Unpacked data center identity in the local community, including needs, benefits, and expectations for future sustainability.
- Developed human-centric guidelines for data center operation and establishment to support community sustainability.
- Connected data center operation with HCI and SHCI discourse.
- **Research Output:** OuluRepo Thesis Link | Md Shafiul Alam, Marianne Kinnula, Dorina Rajanen. **Greenwashing is Cheap, Show Me the Numbers: Exploring Community Sustainability Requirements for Data Center Operation.** [in preparation].

Independent Work

CS students' and AI Sustainability

- Collaboration with researchers from Bangladesh on AI sustainability in a low-resource environment.
- Conducted an exploratory search on undergraduate Computer Science students' understanding of AI sustainability.
- Unpacked their learning method related to AI and sustainability.
- Advocated for the inclusion of sustainability topics in AI classes.
- Formulated curriculum update recommendations based on students' practices and expectations for disseminating the AI sustainability concept.
- **Research Output:** Syed Tangim Pasha*, Md Shafiul Alam*, Md. Saad Bin Kamal*, Nusrat Jahan Nishat, Monjurul Ahsan Bhuiyan, Jannatun Noor.**Understanding Computer Science Students' Knowledge and Perceptions of AI Sustainability in Bangladesh.** [Under Review] | Pre-print.

* = first author

Skills

User Research	Participatory Design, Interviews, Think-aloud, Observation, Surveys, Thematic Analysis
UI Design	Figma, Low-fidelity Prototype, High-Fidelity Prototype, Wireframing
Programming	C, Java, Python, Assembly, JavaScript, PHP, HTML, CSS, XML, SQL, UML
Development	Unity, Visual Studio, Codeblocks, NetBeans, Android Studio
Library	NumPy, Pandas, Matplotlib, Scikit-Learn, OpenGL
Tools	GitHub, Zotero, LaTeX, IBM SPSS, Mplus, XAMPP, Tinkercad, Emu8086, Google Colab, Cisco Packet Tracer
Hardware	Arduino
Language	Bengali [first language], English: C1 [IELTS: 4th Dec, 2023], Finnish: A1.3 [10 ECTS, University of Oulu course], Hindi [fluency in listening and speaking]

Notable Academic Projects

Details and Documentation of all the work can be found in the **Portfolio:**

Design and User Studies

- **Ecoland: Econews:** Designed, developed, and documented a sustainability game focused on climate change, environmental awareness, and environmental news annotation. | **Details**
- **XR Meditation:** Usability, user experience, and research data collection of a VR game. | **Details**
- **DigiPet:** User experience design and user study of a retro game. | **Details**
- **Securo Online:** Behavior change support system with a persuasive system framework for online privacy and security behavior change. | **Details**
- **Multi-National Interact:** Design and User study of an immigrant community app. | **Details**
- **Residente Attivo:** Design of a climate change, Healthy Lifestyle, and Sustainability mobile game for the urban population. | **Details**

Development Works

- **Real-Time Password-Protected System:** Arduino based password-protected system. | **Details**
- **Eye's Suhrid:** An Eye well-being website. | **Details**
- **Onubadok:** Road sign translator Android app. | **Details**
- **Food Value Generator:** Assembly language based calorie calculator app. | **Details**

Teaching Experience

Fall 20 - Spring 21	**Lab Prefect**, CSE 232: Microprocessor, Embedded Systems & IOT Lab, Daffodil International University (120 students).

Awards & Achievement

- **University of Oulu Scholarship** - M.S. Tuition Fee Waiver (2024-2026)
- **Tuition Fee Scholarship** - B.S. Tuition Fee Waiver, Daffodil International University (2017-2022)
- **IT Quiz Competition** - 1st Position, Daffodil ICT Carnival (2018)

Volunteer Contribution

- Information Processing Science thesis template updated to the latest version in Overleaf. (2026) | **Template Link**

Advisee

Undergraduate Students

- **Ifat Bin Amin** — B.S. in Electrical and Electronics Engineering, North South University, Dhaka, Bangladesh, 2025-Present